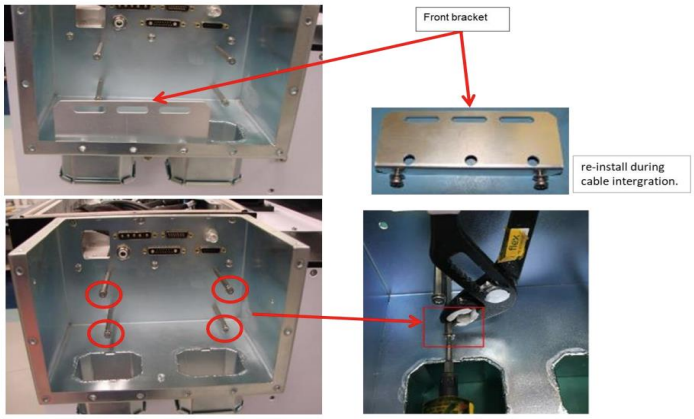
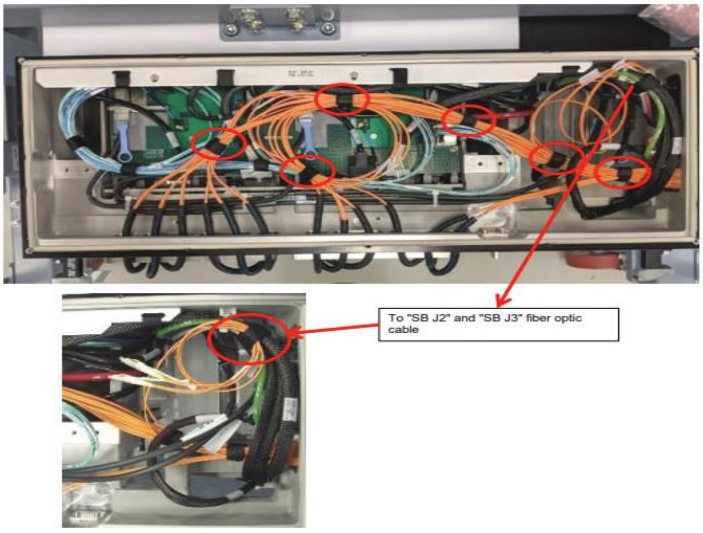
Build Overview:

1 10G Cable Install





CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	653-634-00	ALUMINUM 10G CABLE BRACKET BRACKET OR MISC ARTI	1
2	654-455-00	CN,FIBER OPT,45DEG LC STACKABLE ADAPTER 5POS,3 R	1
3	470-500-01	SCREW,METRIC PH, M2X.4X6MM	4
4	650-412-00	CABLE,UF2 12SLOT LC_LC SUPPBRD TO ENET,OPTICAL FIB	2
5	638-930-01	CABLE, UF2 INST FO SHORT, TELECOM, WITH CONN	2
6	638-930-02	CABLE, UF2 INST FO SHORT, OPTICAL FIBER	7
7	638-930-03	CABLE, UF2 INST FO SHORT, OPTICAL FIBER	3
8	650-609-00	RUBBER, CABLE PLUG, OTHER ARTICLES	12
9	640-834-01	UOM:EA=mm(drawing defines),PLASTIC, UF2 INST PROTECTIVE SLEEVE, HOSE, PIPE OR TUBE	12
10	813-929-00	HDW VELCRO STRP 5/8	9
11	202-591-06	SCRW PAN HEAD TORX M4X.7P/6MM SS W/NYLOK	2
12	938-195-00	WI TIE MOUNT #10 SCREW NATURAL	2
13	710-048-10	TRANCEIVERS PIN SFP+ MULTI 10GBS HOTPLUG DUPLEX	2
14	813-929-01	HDW VELCRO STRP 5/8	6

METHOD:

Flow chart of operation steps

BPM preparation

Preparation Tools

Prepare Materials

Complete Operation

Inventory Materials

Operation Check

Inventory tool

TOOL NAME	TORQUE VALUE
Diagonal pliers	NA
PH1 Phillips screwdriver	0.4 N•m
PH2 Phillips screwdriver	1.6 N•m
T20 Torx wrench	1.6 N·m
PH2 Phillips torque screwdriver	3.3 N·m
Tool (654-776-00)	NA
6mm open end	NA

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

Diagonal pliers, Open end wrench 6mm

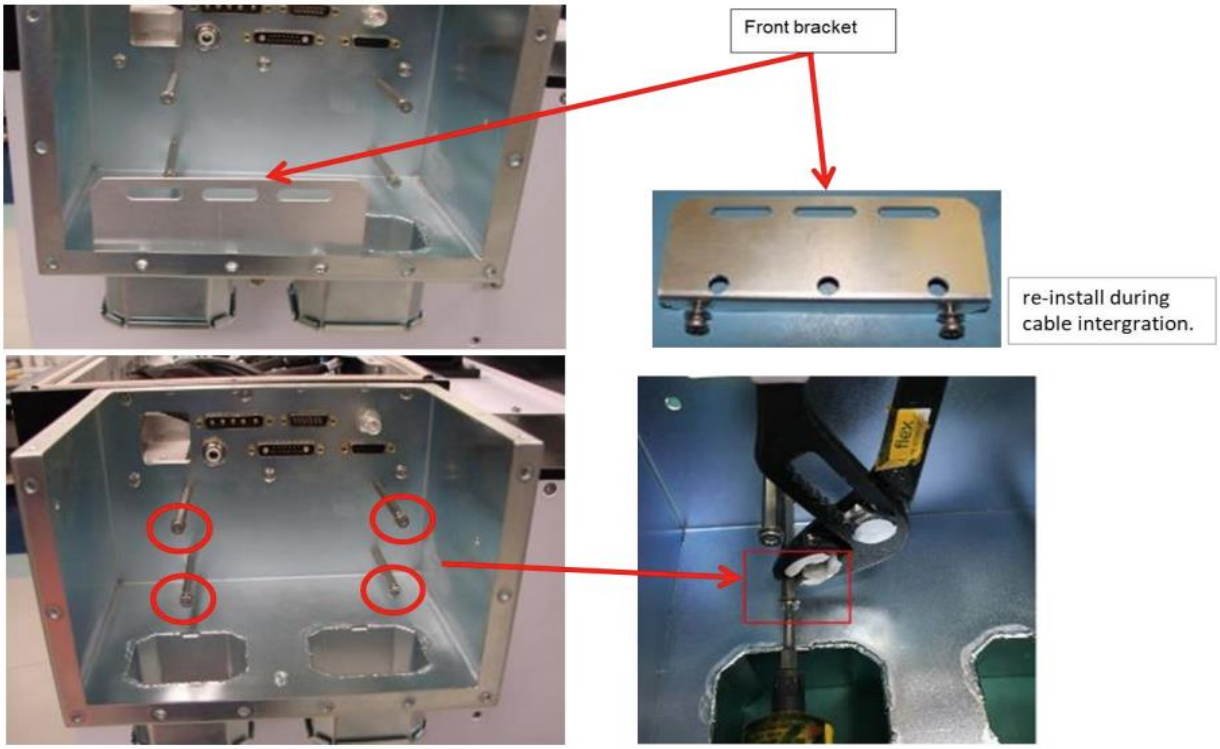
METHOD:

Operation steps

1.1 Remove the front bracket from the disconnection panel (dog house) and keep it together with the screws for later reinstallation.

1.2 Remove the 4 screws and washers from the panel studs, clamp the studs with diagonal pliers or open end wrench 6mm then remove the screws, and save the 4 screws together with the items in step 1.

Note: When removing the screws, make sure that the clamping studs of the diagonal pliers are not loose!



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	653-634-00	ALUMINUM 10G CABLE BRACKET BRACKET OR MISC ARTICLE	1
2	654-455-00	CN,FIBER OPT,45DEG LC STACKABLE ADAPTER 5POS,3 ROWS,BL	1
3	470-500-01	SCREW,METRIC PH, M2X.4X6MM	4

MACHINE:

N/A

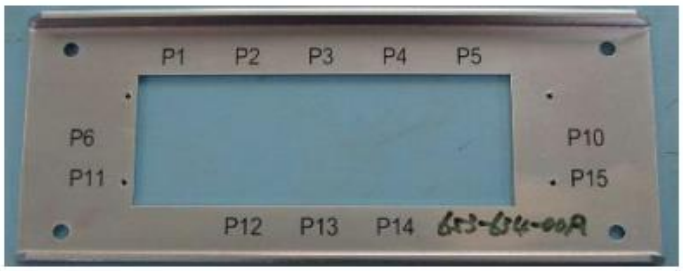
METHOD:

2.1 Take the following materials: i. 1pcs 10G cable support (653-634-00)

pay attention to check that both sides of the support have silk screens printed on the cabinet side and the TH side;

ii. 1pcs 45º fiber connector adapter (654-455-00);

iii. 4pcs M2 screws (470-500-01).



Optical fiber connector adapter

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE:

PH1 Phillips screwdriver 0.4 N•m

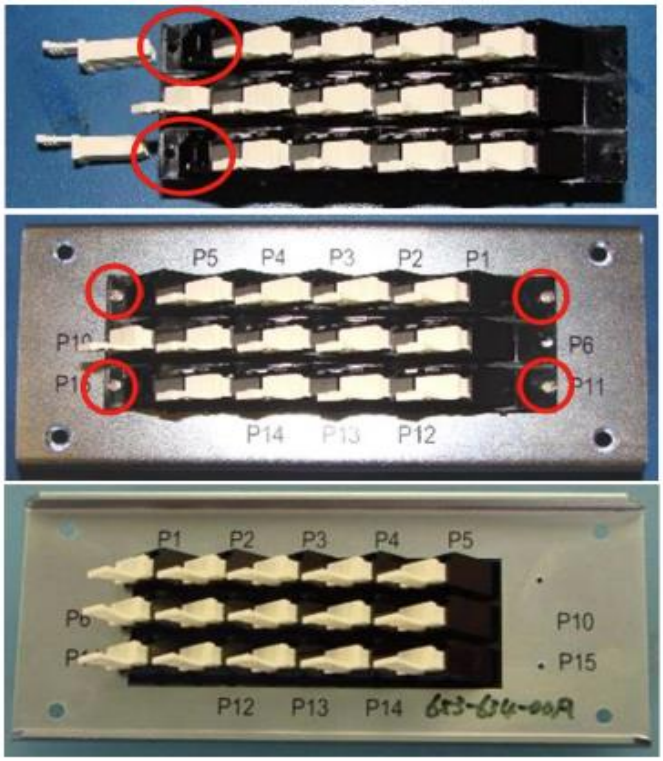
METHOD:

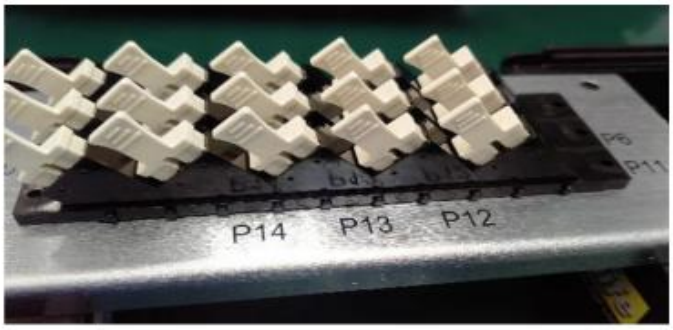
2.2 Remove the two dust plugs from the adapter position shown below.

2.3 Use M2 screws (470-500-01) as shown in the figure to install the adapter on the front of the bracket. The tool is a PH1 Phillips screwdriver and the torque is 0.4 N•m.

2.4 Reinstall the dust plug removed in step 2.2.

2.5 Turn the assembly to the TH side of the bracket.





The embossed edge of the 45° optical fiber adaptor is placed on P14, P13, P12 silk screen of metal bracket cabinet surface.

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
4	650-412-00	CABLE,UF2 12SLOT LC_LC SUPPBRD TO ENET,OPTICAL FIBER	2

MACHINE:

N/A

METHOD:

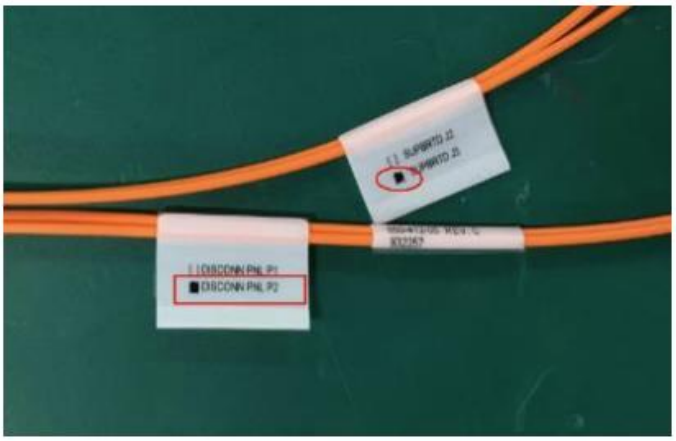
2.6 Take two (650-412-00) optical fibers as shown in the figure.

2.7 The order of labeling of optical fiber cables is as follows:

a. Mark "P1" on the side of the disconnection panel and "J2" on the side of the backplane;

b. Mark "P2" on the side of the disconnection panel and "J3" on the side of the backplane;

2.8 Put these two optical fiber cables aside first and use them later.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
5	638-930-01	CABLE, UF2 INST FO SHORT, TELECOM, WITH CONN	2
6	638-930-02	CABLE, UF2 INST FO SHORT, OPTICAL FIBER	7
7	638-930-03	CABLE, UF2 INST FO SHORT, OPTICAL FIBER	3

MACHINE:

N/A

METHOD:

2.9 Take the following 3 types of optical fiber cables:

Check optical fiber cables with visual fault locator kits.

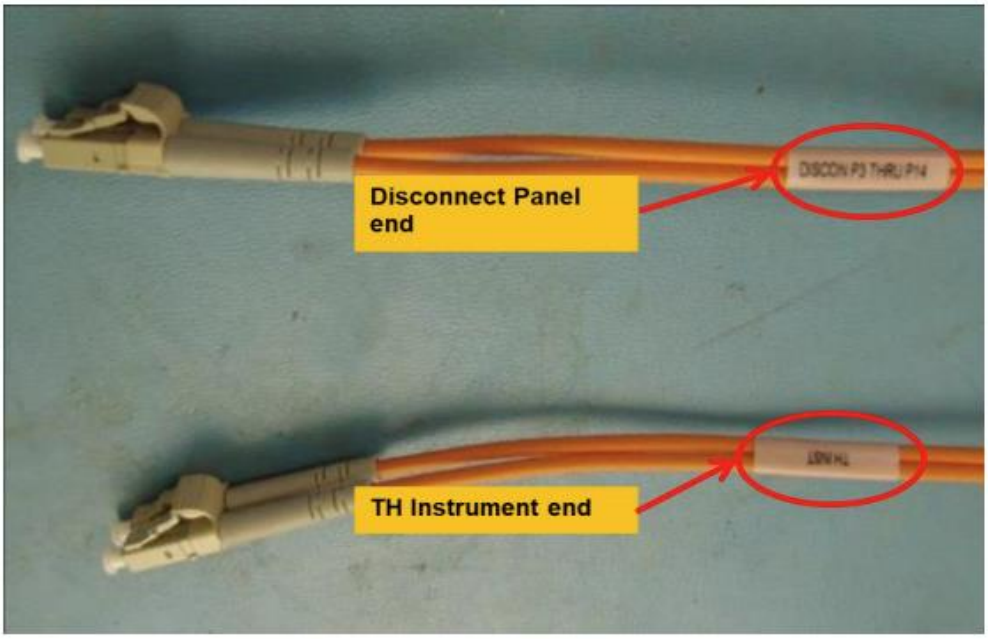
a. 2x UF2 Inst Short, Telecom, With Conn (638-930-01);

b. 7x UF2 Inst Short, Optical Fiber (638-930-02);

c. 3x UF2 Inst Short, Optical Fiber (638-930-03);

2.10 Note that the optical fiber cable has "TH Instrument" and "Disconnect Panel" at both ends, don't insert it wrong when used later

Warning: The optical cable is very sensitive to damage and must not be bent, twisted, clamped, dropped or stepped on!



Safety Tips: N/A

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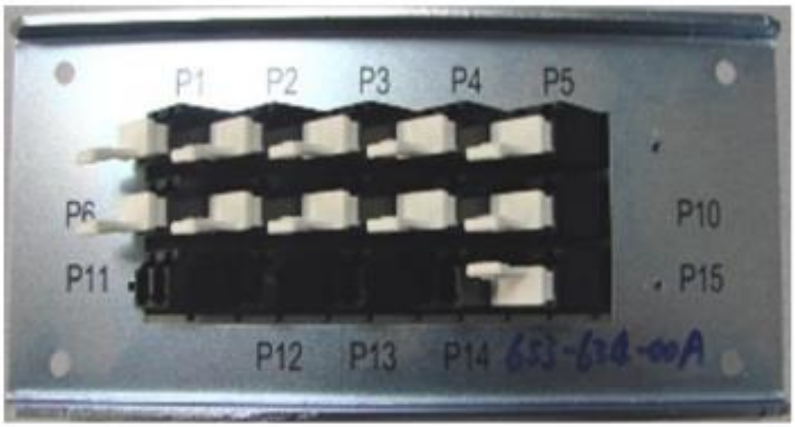
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

3.1 Remove the dust cover from the position "P11" to "P14" on the rear side of the fiber optic adapter, and the dust cover on "P15" remains.



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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

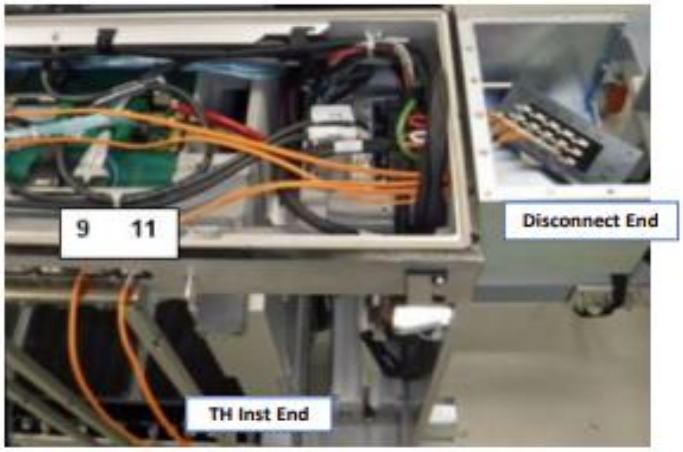
METHOD:

3.2 Unplug the dust cover on the first set of optical fiber ends, and insert the adapters "P14" to "P11" according to the P/N of the optical fiber, as shown in the following table.

3.3 Route the first two optical fibers from P14 and P13 to slot 11 and slot 9, respectively.

DISCONNECT end inserted into the 45º optical fiber adaptor.

3.4 Repeat the following two cables according to the method steps of 3.3 (P12->7 slot, P11->5 slot)



Group 1 (1st row)					
P (#)	P15	P14	P13	P12	P11
CABLE P/N	OPEN	638-930-01	638-930-01	638-930-02	638-930-02
SLOT (#)	N/A	11	9	7	5

TERADYNE PHILIPPINES LTD. _ CEBU

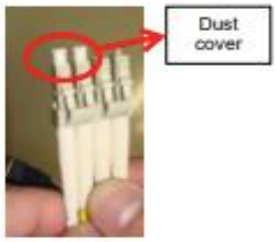
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

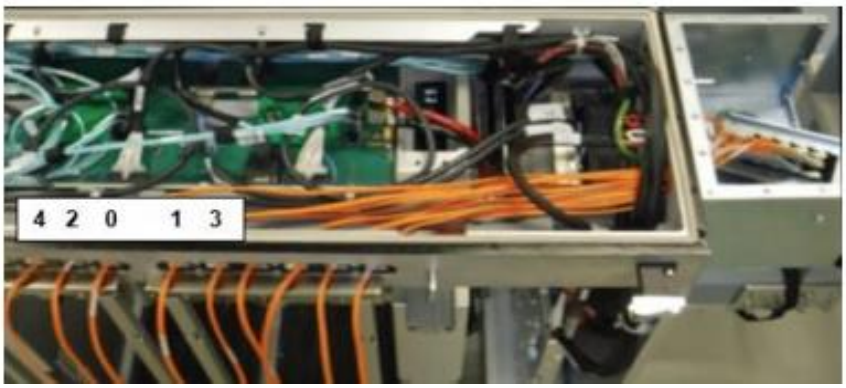
4.1 Remove the dust cover from the position "P6" to "P10" on the rear side of the fiber optic adapter.



4.2 Remove the dust cover on the second set of fiber optic ends, and insert the "P6" to "P10" adapter according to the fiber optic P/N, as shown in the following table

4.3 Route the first 2 optical fibers from P10 and P9 to Slot 3 and Slot 1, respectively.

4.4 Repeat the following 3 optical fibers according to the 4.3 method steps (P8->0 slot, P7->2 slot, P6->4 slot)



Group 2 (2nd row)					
P (#)	P10	P9	P8	P7	P6
CABLE P/N	638-930-02	638-930-02	638-930-02	638-930-02	638-930-02
SLOT (#)	3	1	0	2	4

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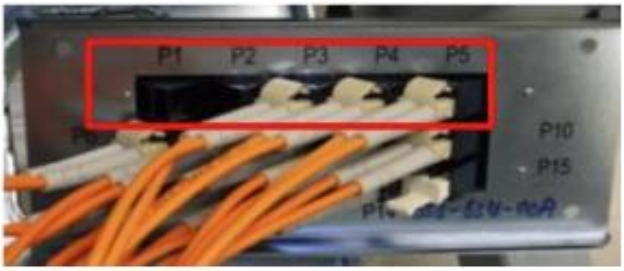
MAN: CERTIFIED OPERATOR, TECHNICIAN

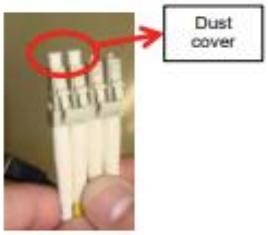
MATERIAL: N/A

MACHINE: N/A

METHOD:

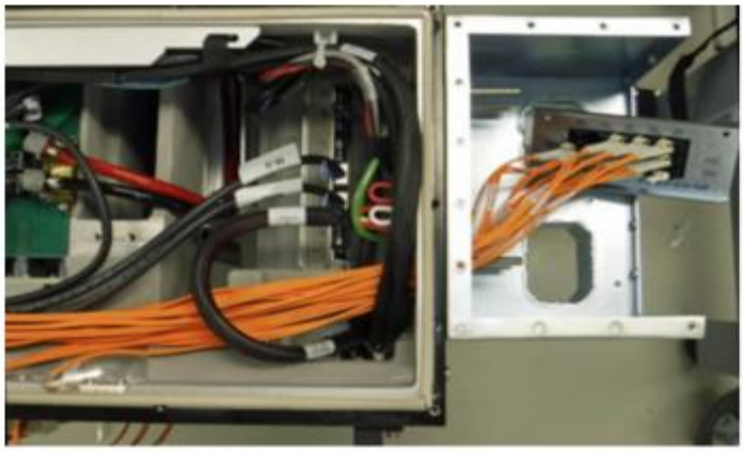
5.1 Remove the dust cover from the position "P1" to "P5" on the rear side of the fiber optic adapter.





5.2 Unplug the dust cover on the third group of optical fiber ends, and insert the "P5" to "P3" of the adapter according to the P/N of the optical fiber, as shown in the following table.

5.3 Route the three fiber cables from P5, P4, and P3 to slot 6, slot 8, and slot 10, respectively.



Group 3 (3rd row)					
P (#)	P5	P4	P3	P2	P1
CABLE P/N	638-930-03	638-930-03	638-930-03	650-412-00	650-412-00
SLOT (#)	6	8	10	SPJ3	SPJ2

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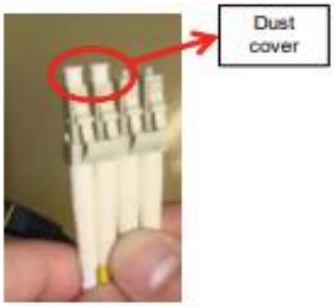
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

6.1 Unplug the dust cover from the two fiber cables 250-412-00, and insert them into the fiber adapter ports in the following order.
A."DISCONN PNL P1"Plug in"P1"
B."DISCONN PNL P2"Insert"P2"



6.2 Route the two fiber cables from P1 and P2 to the upper backplane area.

Group 3 (3rd row)					
P (#)	P5	P4	P3	P2	P1
CABLE P/N	638-030-03	638-030-03	638-030-03	650-412-00	650-412-00
SLOT (#)	6	8	10	SPJ3	SPJ2

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

7.1 Use the four M4 screws retained before to connect the wire bracket assembly to the four studs as shown in the figure. The tool is a PH2 Phillips screwdriver with a torque of 1.6 N•m.

Pay attention to the alignment of fiber cables behind the panel. Warning: The optical cable is very sensitive to damage and must not be bent, twisted, clamped, dropped or stepped on!

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

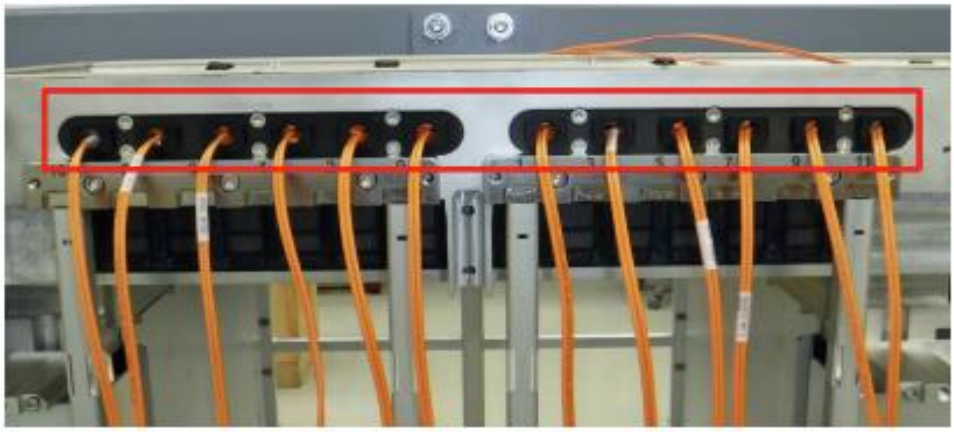
ITEM	PART NUMBER	DESCRIPTION	QTY
8	650-609-00	RUBBER, CABLE PLUG, OTHER ARTICLES	12

MACHINE:

N/A

METHOD:

8.1 As shown in the figure, put 12 cable plugs (650-609-00) on the 12 optical fiber lines on the 12 hole positions of the panel bracket, a total of 12.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
9	640-834-01	UOM:EA=mm(drawing defines),PLASTIC,UF2 INST PROTECTIVE SLRUBBER, CABLE PLUG, OTHER ARTICLES	12

MACHINE:

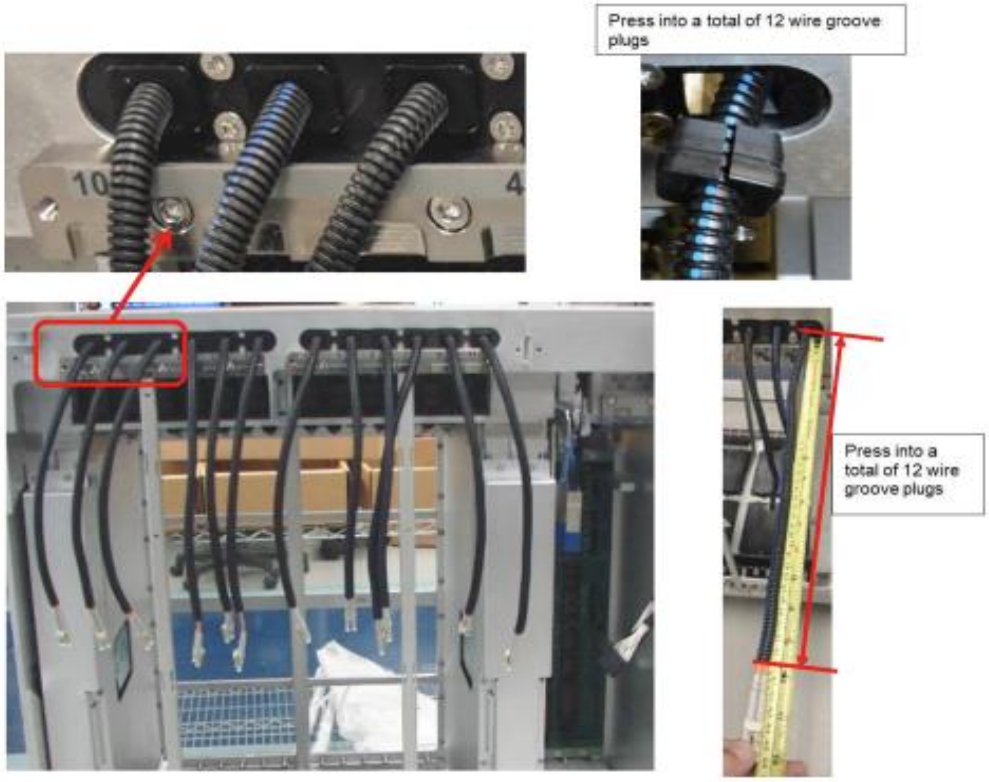
N/A

METHOD:

8.2 Use the tool (654-776-00) as shown in the figure to install a black protective tube (640-834-01) on the optical fiber line. Each optical fiber has a protective tube, a total of 12 tubes.

8.3 Press the protective tube into the TH slot plug.

8.4 As shown in the figure, adjust the length of the optical fiber protection tube outside the frame from the cable slot to the end of the optical fiber end to 16 inches.



Safety Tips: N/A

TQC: N/A

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

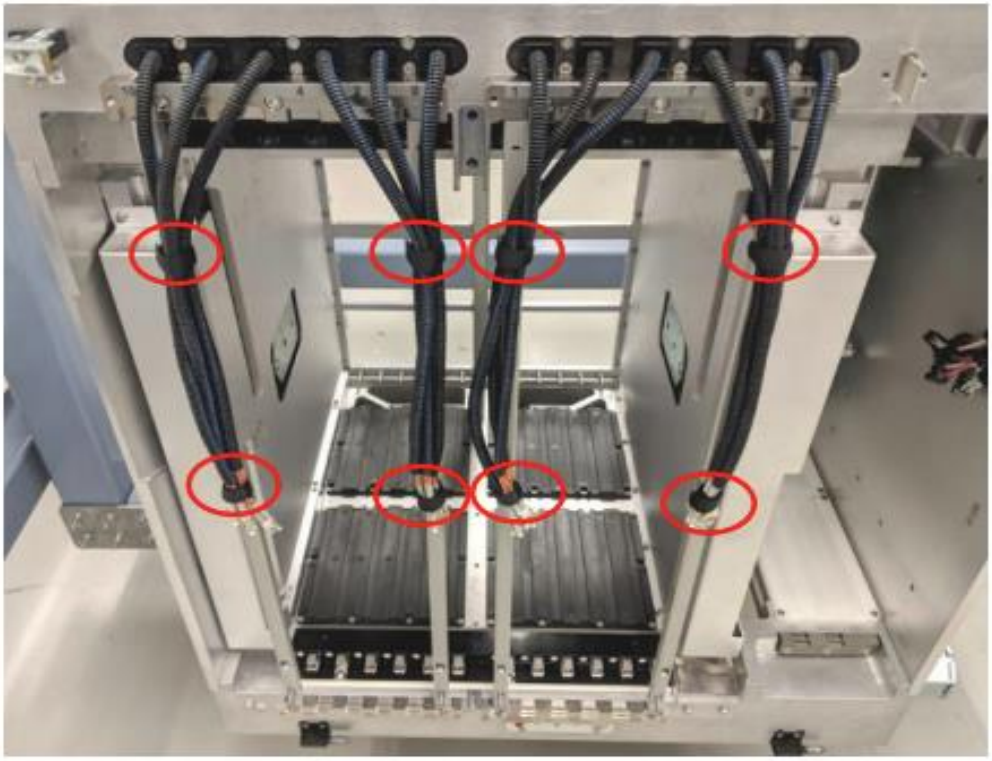
ITEM	PART NUMBER	DESCRIPTION	QTY
10	813-929-00	HDW VELCRO STRP 5/8	8

MACHINE:

N/A

METHOD:

8.5 As shown in the figure, use eight cable ties (813-929-00) to bundle the even-numbered and odd-numbered optical fibers on the cable bracket, two for each bracket.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION				QTY
11	202-591-06	SCRW PAN HEAD TORX M4X.7P/6MM SS W/NYLOK				2
12	938-195-00	WI TIE MOUNT #10 SCREW NATURAL				2

MACHINE:

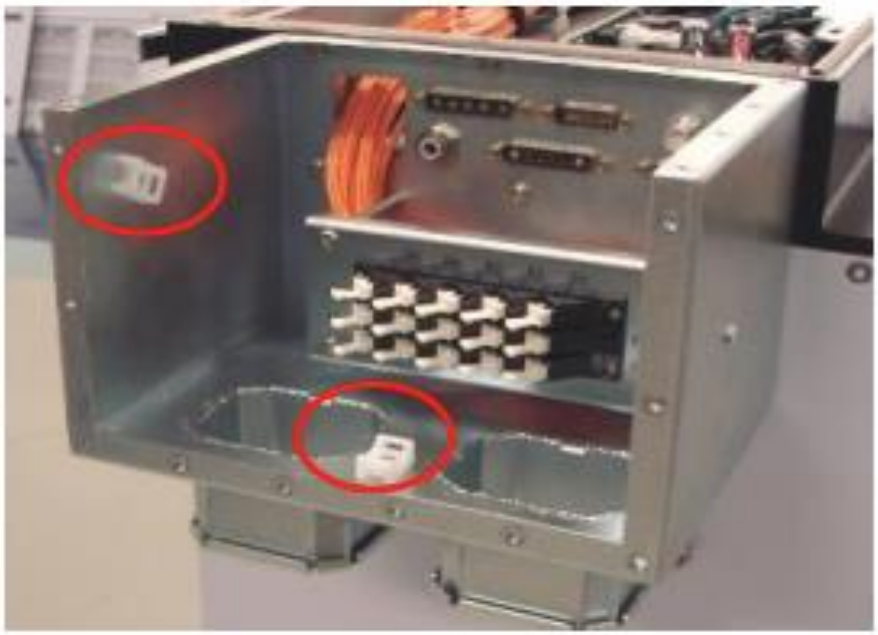
T20 Torx wrench 1.6 N·m, PH2 Phillips torque screwdriver 3.3 N·m

METHOD:

9.1 Use two M4 screws (202-591-06) to install two (938-195-00) cable connection seats in the position shown in the panel.

Pay attention to the direction of installation.

The tool is a T20 Torx wrench with a torque value of 1.6 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

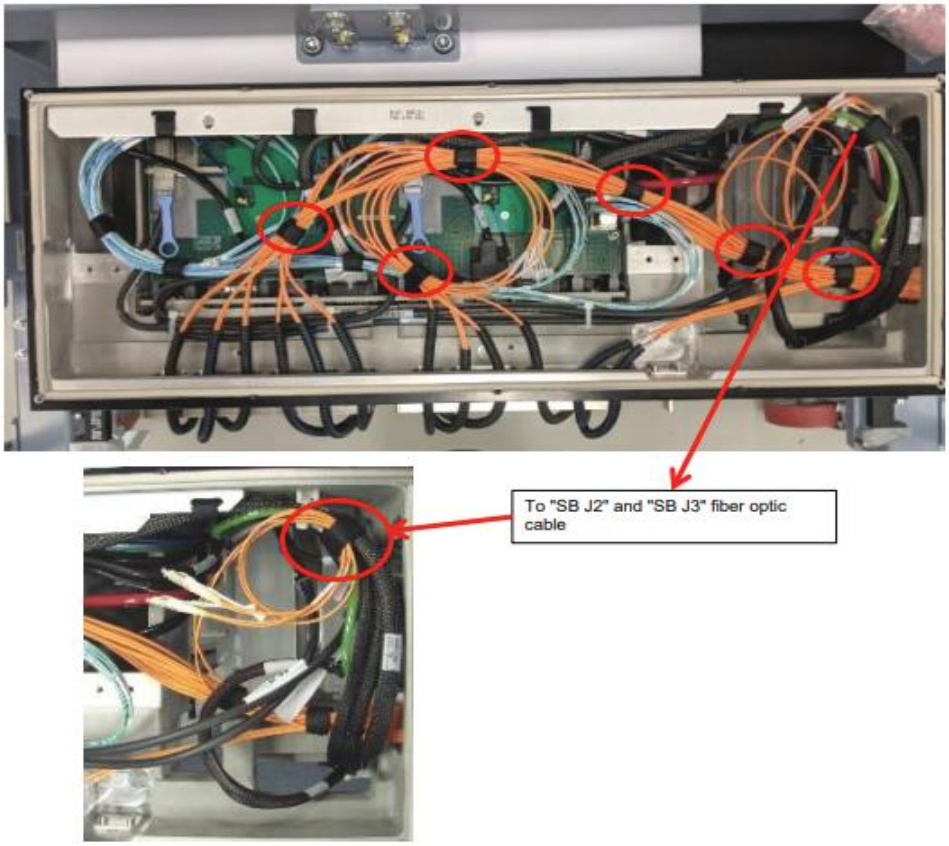
ITEM	PART NUMBER	DESCRIPTION	QTY
13	813-929-01	HDW VELCRO STRP 5/8	6

MACHINE:

N/A

METHOD:

10.1 As shown in the figure, use six cable ties (813-929-01) to organize the optical fibers in the frame.
10.2 Pay attention to the two 650-412-00 fiber cables marked "J2" and "J3" on the backplane side, and use the existing cable ties to temporarily tie the coil to the main cable bundle.



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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION				QTY
14	710-048-10	TRANCEIVERS PIN SFP+ MULTI 10GBS HOTPLUG DUPLEX CN 850NM				2
15	813-929-00	HDW VELCRO STRP 5/8				1

MACHINE:

N/A

METHOD:

10.3 Take two transceivers (710-048-10), do not open the package, tie the package with a cable tie (813-929-00) to the TH bracket.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Matters needing attention

1. Do ESD and protective safety measures.
2. Be careful during the assembly process so as not to scratch the surface of the parts.
3. After assembly, check the assembled parts for defects and leak-free operation.

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